



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

AI-Based Sustainable Agriculture Management System

A CAPSTONE PROJECT REPORT

*A Capstone Project Report submitted in partial fulfilment of the requirements for the
Course of*

CSA0923 - PROGRAMMING IN JAVA

to the award of the degree of

BACHELOR OF ENGINEERING / BACHELOR OF TECHNOLOGY IN

Computer Science and Engineering (Artificial Engineering)

Submitted by

Y. Divya Sree (192472357)

Under the Supervision of

Dr. Sarala V

SIMATS ENGINEERING

Saveetha Institute of Medical and Technical Sciences

Chennai-602105

SEPTEMBER 2026

SIMATS ENGINEERING

Saveetha Institute of Medical and Technical Sciences

Chennai-602105

BONAFIDE CERTIFICATE

This is to certify that the Capstone Project entitled **“AI-Based Sustainable Agriculture Management System”** has been carried out by **Y. Divya Sree** under the supervision of Dr. **Sarala V** and is submitted in partial fulfilment of the requirements for the current semester of the B. Tech Computer Science and Engineering program at Saveetha Institute of Medical and Technical Sciences, Chennai.

COURSE COORDINATOR

Dr. G. Charlyn Pushpa Latha

Professor

Department of CSE

Saveetha School of Engineering,
SIMATS, Chennai – 602105.

COURSE FACULTY

Dr. Sarala V

Professor

Department of CSE

Saveetha School of Engineering,
SIMATS, Chennai – 602105.

SIMATS ENGINEERING

Saveetha Institute of Medical and Technical Sciences

Chennai-602105

DECLARATION

We, **Y. Divya Sree** of the Computer Science and Engineering, SIMATS Engineering, Saveetha Institute of Medical and Technical Sciences, Chennai, hereby declare that the Capstone Project Work entitled “**AI-Based Sustainable Agriculture Management System**” is the result of our own Bonafide efforts. To the best of our knowledge, the work presented herein is original, accurate, and has been carried out in accordance with the principles of engineering ethics and academic integrity. All sources, references, data, and other materials used in the project have been appropriately acknowledged. We further declare that the data, results, analysis, and findings presented in this report have not been fabricated, falsified, or manipulated. Any external tools, software, datasets, computational resources, or AI-assisted tools used during the project have been appropriately disclosed. We confirm that all applicable ethical, academic, institutional, and professional requirements have been duly followed throughout the planning, execution, analysis, and documentation of the project.

Place: Chennai

Date: 08-09-2026

Y. Divya Sree (192472357)

SIMATS ENGINEERING

Saveetha Institute of Medical and Technical Sciences

Chennai-602105

INDIVIDUAL CONTRIBUTION STATEMENT

Student Name / Register No.	Specific Responsibilities	Design & Development Contribution	Testing & Analysis Contribution	Report Contribution	Approx . Contrib ution (%)
R. Geethika / 1924251 01	Module 1 – Priority-Based Farm Task Management	Designed and implemented the priority-based task management module using priority queue logic	Tested task prioritization and selection of high-priority tasks	Contributed to report preparation and documentation	33%
B. Nikhila / 1924721 02	Module 3 – Task Completion and Performance Monitoring	Designed and implemented task completion and performance monitoring functionality	Tested task status updates, completed/pending task counting, and performance monitoring	Contributed to report preparation and documentation	33%
Y. Divya Sree / 1924723 57	Module 2 – Farm Condition Analysis; Website Design & Development; PPT Preparation and Presentation Support	Designed and developed the website and implemented Module 2 using conditional decision logic	Tested farm data entry, condition analysis, task recommendations, and result display	Contributed to report preparation, PPT preparation, and presentation	34%

ABSTRACT

Agriculture involves several decisions related to crop selection, soil conditions, water management, crop health, and weather conditions. Making these decisions manually can be difficult because multiple farm conditions need to be considered together. The project “AI-Based Sustainable Agriculture Management System” proposes a software-based decision-support system that helps organize farm information and generate suitable farm-management tasks based on the entered conditions.

The proposed system accepts farm-related inputs such as crop, soil type, soil moisture, weather condition, and crop health. The system uses three main modules. Module 1, Priority-Based Farm Task Management, uses a Priority Queue to arrange tasks according to their importance, with high-priority tasks handled before medium- and low-priority tasks. Module 2, Conditional Decision Logic-Based Farm Condition Analysis, uses IF-ELSE decision logic to analyse the entered farm conditions. For example, low soil moisture generates an irrigation task with high priority, while poor crop health generates a disease-monitoring task. Rainy weather can also generate a disease-monitoring task. Module 3, Task Completion and Performance Monitoring, tracks task status and distinguishes between pending and completed tasks.

The system is implemented using Java-based programming concepts with a web interface. Farm and task information can be stored and retrieved through the system so that the dashboard reflects user entered information rather than fabricated values. The project focuses on providing a simple and understandable decision-support workflow while encouraging efficient use of agricultural resources. The implemented system demonstrates how basic AI-oriented decision support, priority-based task management, and task monitoring can be combined into a single sustainable agriculture management application. The present prototype is primarily based on defined decision rules and does not claim trained machine-learning accuracy. Future development can integrate real-time IoT sensors, weather APIs, machine-learning models, disease-image analysis, and advanced analytics.

Keywords: Sustainable Agriculture, Decision Support System, Priority Queue, IF-ELSE Logic, Farm Management, Artificial Intelligence.

TABLE OF CONTENTS

Sl. No	Title	Page No
i	CERTIFICATE FROM THE SUPERVISOR	i
ii	DECLARATION BY THE CANDIDATE	ii
iii	INDIVIDUAL CONTRIBUTION STATEMENT	iii
iv	ABSTRACT	iv
v	LIST OF FIGURES	v
vi	LIST OF TABLES	vi
vii	LIST OF ABBREVIATIONS / SYMBOLS	vii
1	CHAPTER 1: INTRODUCTION AND ENGINEERING PROBLEM	10–12
2	CHAPTER 2: LITERATURE REVIEW AND EXISTING SOLUTIONS	13–14
3	CHAPTER 3: ENGINEERING DESIGN & TRADE-OFFS, SUSTAINABILITY, ETHICS, SAFETY, RISK & SECURITY	15–20
4	CHAPTER 4: IMPLEMENTATION, TESTING & RESULTS, PROJECT MANAGEMENT	21-25
5	CHAPTER 5: CONCLUSION, FUTURE WORK AND REFLECTION	26- 27
	REFERENCES	28
	APPENDICES	29-32

LIST OF FIGURES

Figure No.	Figure Name	Page No.
Figure 1	System Architecture and Workflow	27
Figure 2	Website / Home Dashboard	18
Figure 3	Farm Data Input Interface	19
Figure 4	Module 2 Farm Condition Analysis Output	20
Figure 5	Priority-Based Task Management Output	21
Figure 6	Task Completion and Performance Dashboard	22

LIST OF TABLES

Table No.	Table Name	Page No.
Table 1	Functional Requirements	15
Table 2	Design Specifications	16
Table 3	Alternative Solutions Evaluation	16
Table 4	Sustainability, Ethics, Safety, Risk and Security Analysis	17
Table 5	Risk Register	17
Table 6	Software Resources Used	18
Table 7	Software Tools and Modern Technologies Used	19
Table 8	Achievement of Project Objectives	21
Table 9	Project Milestones	22
Table 10	Team Roles and Contributions	23

LIST OF ABBREVIATIONS / SYMBOLS

Abbreviation / Symbol	Description
AI	Artificial Intelligence
IoT	Internet of Things
UI	User Interface
DB	Database
IF-ELSE	Conditional Decision Logic
PQ	Priority Queue

CHAPTER 1

INTRODUCTION AND ENGINEERING PROBLEM

1.1 Background

Agriculture is an important sector that depends on several interacting factors such as soil condition, water availability, weather, crop type, and crop health. Farmers need to make timely decisions regarding irrigation, crop monitoring, and other farm activities. When these decisions are managed manually, it may become difficult to organize tasks and respond to changing farm conditions efficiently.

The development of digital agriculture systems provides an opportunity to support farmers through data-based decision making. An agriculture management system can collect relevant farm information, analyse the entered conditions, and provide appropriate recommendations or tasks. The proposed **AI-Based Sustainable Agriculture Management System** is designed as a softwarebased decision-support application. It combines farm-condition analysis, priority-based task management, and task completion monitoring. The system uses defined decision rules to analyse farm conditions and a Priority Queue to ensure that important tasks are addressed first.

1.2 Need for the Project

The project is needed because agricultural decisions often involve multiple conditions at the same time. In particular, soil moisture, weather, and crop health can affect the actions required on a farm.

The major needs addressed by the project are:

- To simplify farm-condition analysis.
- To identify important farm tasks from entered conditions.
- To prioritize urgent tasks.
- To reduce unnecessary or unplanned resource usage.
- To maintain task status in an organized manner.
- To provide a simple dashboard for monitoring farm activities.
- To support sustainable use of agricultural resources.

Farmers and farm managers are the primary users who can benefit from organized decision support. From an engineering perspective, the project demonstrates the integration of decision logic, data management, task prioritization, and monitoring into one software system.

1.3 Problem Statement

Farm management requires decisions based on multiple interacting conditions such as crop type, soil type, soil moisture, weather, and crop health. Manual analysis of these conditions can make it difficult to identify urgent tasks and monitor their completion.

Therefore, the engineering problem is to develop a software-based agriculture management system that accepts farm-condition information, analyses it using defined decision logic, generates appropriate tasks, prioritizes important tasks, and tracks task completion without using fabricated farm data.

1.4 Project Aim

The aim of this project is to develop an **AI-Based Sustainable Agriculture Management System** that supports farm-condition analysis, task prioritization, and performance monitoring using softwarebased decision-support techniques.

1.5 Project Objectives

1. To design a software system for organizing agricultural information.
2. To develop a web-based interface for entering and viewing farm information.
3. To analyse farm conditions using IF-ELSE decision logic.
4. To generate suitable farm-management tasks based on entered conditions.
5. To prioritize generated tasks using a Priority Queue.
6. To track pending and completed farm tasks.
7. To provide a dashboard for monitoring task status.
8. To avoid fabricated values and display information based on actual user-entered data.
9. To support sustainable decision making related to agricultural resource management.

1.6 Scope Included

- Farm information entry.
- Crop, soil, moisture, weather, and crop-health analysis.
- Rule-based farm-condition decision support.
- Priority-based task management.
- Task status tracking.
- Dashboard-based monitoring.
- Storage and retrieval of entered farm/task information.

Excluded

- Real-time agricultural sensor hardware.
- Automatic physical irrigation control.
- Actual pesticide or fertilizer application.
- Guaranteed disease diagnosis.
- A trained machine-learning model unless separately integrated.
- Replacement of professional agricultural advice.

Assumptions

- Users enter meaningful and valid farm information.
- The decision rules are based on the conditions defined for the project.
- The system operates using the available software environment.

Boundary Conditions

The system provides software-based decision support. It does not directly control agricultural equipment or guarantee crop outcomes.

1.7 Expected Engineering Outcome

The expected outcome is a functional prototype of an agriculture management system that:

- Accepts farm-condition information.
- Analyses the entered information.
- Generates relevant farm tasks.
- Assigns priority to important tasks.
- Tracks task completion.
- Displays the current task status.
- Stores actual user-entered information.
- Provides a foundation for future IoT and machine-learning integration.

CHAPTER 2

LITERATURE REVIEW AND EXISTING SOLUTIONS

2.1 Review Method

The literature review was carried out using recent research related to artificial intelligence in agriculture, sustainable agriculture, machine learning, IoT-based agricultural systems, and digital agricultural decision support. The reviewed studies were used to understand existing approaches, their benefits, limitations, and the engineering gap addressed by the proposed system.

2.2 Review of Existing Work AI for Sustainable Agriculture

Recent research has investigated the use of Artificial Intelligence for improving agricultural productivity and sustainability. AI-based methods can support crop production, resource management, monitoring, and agricultural decision making.

AI and IoT for Soil Management

AI and IoT-based approaches can combine sensor information with intelligent analysis for soil and nutrient management. Such systems provide opportunities for more precise agricultural resource management.

Machine Learning in Agriculture

Machine-learning techniques have been studied for applications such as crop prediction, disease detection, yield prediction, and agricultural monitoring. These approaches can provide predictive capabilities when suitable datasets and trained models are available.

Digital Agriculture and Decision Support

Digital agricultural systems can assist users by organizing farm data and providing recommendations based on available information. However, system complexity, sensor requirements, data availability, and implementation cost can affect practical adoption.

2.3 Comparative Analysis

Existing Approach	Advantages	Limitations	Relevance to Proposed Work
Manual Farm Management	Simple and familiar	Time-consuming and difficult to prioritize tasks	Problem addressed by proposed system

AI-Based Agriculture	Can support intelligent decision making	May require datasets and trained models	Provides future direction
IoT-Based Agriculture	Provides real-time sensor information	Requires hardware and sensor infrastructure	Considered as future enhancement
Proposed System	Simple decision support, task prioritization and monitoring	Current prototype uses defined decision rules	Directly supports project objectives

2.4 Research / Engineering Gap

Existing agricultural technologies provide different capabilities such as prediction, monitoring, sensor-based data collection, and intelligent analysis. However, a lightweight system that combines farm-condition decision logic, priority-based task management, and task completion monitoring can provide a simpler workflow for academic and prototype-level implementation.

The proposed project addresses this gap by integrating these functions into a single system. The current implementation focuses on rule-based decision support rather than claiming predictive machine-learning performance.

2.5 Proposed Contribution

The proposed system contributes a simple software-based framework for sustainable agriculture management. Its main contribution is the integration of:

- Farm-condition analysis.
- IF-ELSE based decision logic.
- Priority Queue based task management.
- Task completion tracking.
- Dashboard-based performance monitoring.

The system also follows the project requirement of displaying actual user-entered information rather than fabricated results.

CHAPTER 3

ENGINEERING DESIGN, TRADE-OFFS, SUSTAINABILITY, ETHICS, SAFETY, RISK AND SECURITY

3.1 Requirements

3.1.1 Stakeholder Requirements

The primary stakeholder is the farmer or farm user who requires a simple method for understanding farm conditions and organizing agricultural tasks.

The system should:

- Accept basic farm and crop information.
- Analyse entered conditions.
- Generate appropriate farm tasks.
- Identify high-priority activities.
- Allow users to monitor task status.
- Store farm and task information.
- Present information clearly through a web interface.

3.1.2 Functional Requirements

Requirement ID	Functional Requirement
FR1	The system shall accept farm and crop information.
FR2	The system shall accept soil-moisture information.
FR3	The system shall accept weather and crop-health information.
FR4	The system shall analyse entered conditions.
FR5	The system shall generate suitable farm tasks.
FR6	The system shall assign task priorities.
FR7	The system shall display pending and completed tasks.
FR8	The system shall store relevant information in the database.
FR9	The system shall provide performance-monitoring information.

3.1.3 Non-Functional Requirements

- **Usability:** The interface should be simple and understandable.
- **Reliability:** The system should generate consistent results for the same inputs.

- **Performance:** The system should provide analysis without unnecessary delay.
- **Maintainability:** The modular structure should allow individual modules to be modified.
- **Security:** User-entered data should be handled securely.
- **Scalability:** The system should allow future integration of additional agricultural parameters.
- **Data Integrity:** Stored information should remain consistent with user actions.

3.2 Constraints and Applicable Standards

Constraints

- The current system is primarily software-based.
- Real-time IoT sensors are not included in the current implementation.
- The current decision-making module uses predefined conditional rules.
- The quality of recommendations depends on the correctness of user-provided information.
- The system is intended as a decision-support prototype rather than a fully automated agricultural control system.

Applicable Software Practices

The project follows software-quality considerations related to usability, reliability, performance, maintainability, and security. Web-security practices should be considered when handling user input and database operations.

3.3 Design Specifications

Component	Specification
Input	Crop, soil type, soil moisture, weather, crop health
Decision Method	IF-ELSE conditional logic
Task Prioritization	Priority Queue
Priority Levels	HIGH, MEDIUM, LOW
Priority Values	HIGH = 3, MEDIUM = 2, LOW = 1
Task States	PENDING, COMPLETED
Database	Lightweight database for farm/task records
Interface	Web-based interface
Core Language	Java

Output	Recommended and prioritized farm tasks
--------	--

3.4 Alternative Solutions and Evaluation

Three possible approaches were considered.

Alternative 1 – Manual Farm Management

Farmers manually observe farm conditions and decide which activity to perform.

Advantages: Simple and inexpensive.

Limitations: Highly dependent on human observation, difficult to prioritize multiple tasks, and difficult to maintain digital records.

Alternative 2 – Full IoT-Based Smart Agriculture System

Sensors continuously collect soil and environmental information and transmit the data to an application.

Advantages: Real-time data collection and automation.

Limitations: Requires hardware, sensors, connectivity, maintenance, and higher implementation cost.

Alternative 3 – Software-Based Decision Support System

The system accepts farm conditions, analyzes them using decision logic, prioritizes tasks, and tracks completion.

Advantages: Lower complexity, easier implementation, simple user interaction, and suitable for a prototype.

Evaluation

Criterion	Manual System	IoT System	Proposed System
Implementation complexity	Low	High	Moderate
Hardware requirement	Low	High	Low
Cost	Low	High	Low
Task prioritization	Low	Medium/High	High
Data management	Low	High	High
Scalability	Low	High	High
Suitability for prototype	High	Medium	High

The proposed system was selected because it provides a balance between functionality, implementation complexity, cost, and future expandability.

3.5 Selected Design and Detailed Design

The selected design consists of three interconnected modules.

Overall System Flow

User Input → Farm Condition Analysis → Task Generation → Priority Assignment → Task Management → Completion Tracking → Performance Dashboard

Module 1: Priority-Based Farm Task Management

This module uses a **Priority Queue**. Each generated task is assigned a priority value:

- HIGH = 3
- MEDIUM = 2
- LOW = 1

The task with the highest priority is selected before lower-priority tasks. **Module**

2: Conditional Decision Logic-Based Farm Condition Analysis

This module analyses the farm conditions using IF-ELSE logic.

The major rules are:

- If soil moisture is low → generate irrigation task with HIGH priority.
- If crop health is poor → generate disease-monitoring task with HIGH priority.
- If weather is rainy → generate disease-monitoring task.

This module converts input conditions into actionable farm tasks.

Module 3: Task Completion and Performance Monitoring

All generated tasks initially have a **PENDING** status. When the user completes a task, its status changes to **COMPLETED**.

The system can then determine:

- Number of completed tasks.
- Number of pending tasks.
- Number of high-priority pending tasks.
- Overall task status.

3.6 Trade-Off Analysis

The main engineering trade-off was between system simplicity and advanced automation.

A fully automated IoT-based system could provide real-time environmental data but would require additional hardware and communication infrastructure. The proposed software-based approach reduces implementation complexity while still demonstrating decision support and task management. Similarly, predefined IF-ELSE rules are easier to understand and implement than a trained machinelearning model. However, machine-learning models could provide more advanced prediction when sufficient real-world datasets become available.

The selected design therefore prioritizes simplicity, transparency, maintainability, and future extensibility.

3.7 Sustainability, Ethics, Safety, Risk and Security

Domain	Consideration	Evidence/Approach	Impact
Sustainability	Reduce unnecessary resource usage	Condition-based irrigation recommendation	Supports efficient water usage
Sustainability	Improve task organization	Priority-based task management	Supports timely farm activities
Ethics	Responsible recommendations	System acts as decision support	Reduces over-reliance on automation
Data Privacy	Protect farm information	Controlled database access	Reduces unauthorized access
Safety	Avoid automatic physical control	Software recommendations only	Reduces physical equipment risk
Reliability	Consistent decision rules	Defined IF-ELSE conditions	Improves predictable behaviour
Security	Validate user inputs	Input validation and safe database operations	Reduces data-related risks

Risk Register

Risk	Likelihood	Severity	Risk Level	Mitigation	Residual Risk
------	------------	----------	------------	------------	---------------

Incorrect user input	Medium	Medium	Medium	Input validation	Low/Medium
Incorrect rule implementation	Low/Medium	High	Medium	Functional testing	Low
Database failure	Low	Medium	Low/Medium	Proper database handling and backup	Low
Unauthorized data access	Low	High	Medium	Access control and secure queries	Low
Dependence on predefined rules	Medium	Medium	Medium	Future ML integration	Medium
Lack of real-time sensor data	High	Medium	Medium	Future IoT integration	Medium

CHAPTER 4

IMPLEMENTATION, TESTING & RESULTS, PROJECT MANAGEMENT

4.1 Development Approach & Resources

The project was developed using a modular approach. The system requirements were first identified and then divided into farm-condition analysis, task prioritization, and task monitoring modules. The web interface was developed to allow users to enter and view farm information, while the application logic processes the entered conditions and generates tasks. The modules were then integrated into the overall agriculture management workflow.

Resources Used

Resource	Purpose
Java	Application logic and implementation
Web technologies	User interface development
Database	Storage and retrieval of farm/task information
Development environment	Coding and testing
AI-assisted development tools, where used	Development support

4.2 Software Implementation & Modern Tools Used User Interface

The website provides interfaces for entering farm information and viewing generated tasks and task status.

Decision Logic

The system analyses:

- Crop
- Soil type
- Soil moisture
- Weather
- Crop health

The entered values are processed through the defined IF-ELSE rules.

Priority Management

Generated tasks are assigned priority levels and managed using the Priority Queue implemented in Module 1.

Task Monitoring

The task-monitoring module records whether tasks are pending or completed and provides corresponding counts. **Tools**

Tool / Software	Purpose	Stage Used
Java	Application and algorithm implementation	Development
HTML/CSS/JavaScript or implemented web technologies	Website interface	Development
Database	Data storage	Development
Development IDE/environment	Coding and debugging	Development
AI-assisted development tool, if used	Development assistance	Development

Implementation

Evidence:

[Insert actual website screenshots here] **Suggested**

figures to insert:

- Figure 1: Home/Dashboard page
- Figure 2: Farm information input page
- Figure 3: Module 2 condition analysis output
- Figure 4: Priority-based task list
- Figure 5: Task completion dashboard

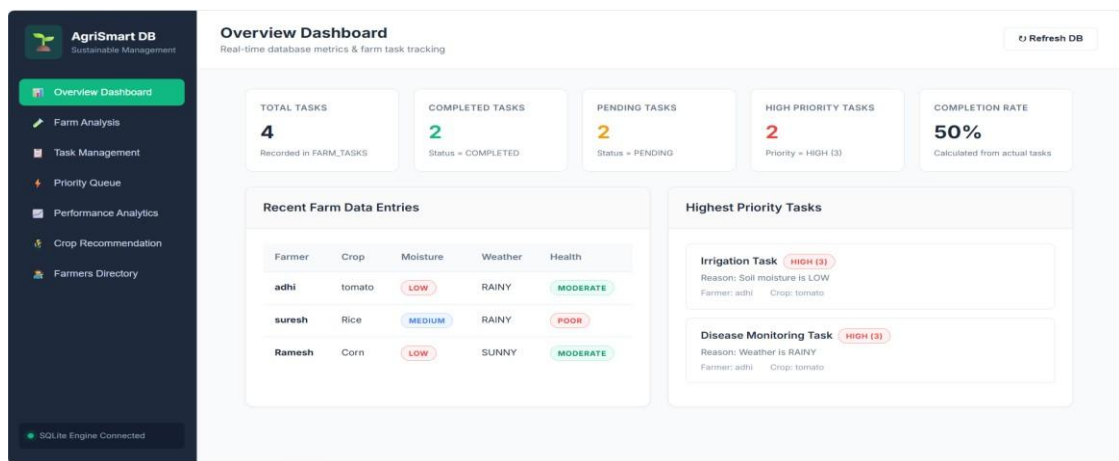
4.4 Results

The implemented system is designed to generate outputs based on actual farm information entered by the user. The system does not require fabricated farm values for displaying dashboard information.

The main functional outputs include:

1. Farm-condition analysis.
2. Irrigation task generation when the defined low-moisture condition occurs.
3. Disease-monitoring task generation for the defined crop-health and weather conditions.
4. Priority-based ordering of generated tasks.
5. Pending and completed task tracking.
6. Dashboard-based task monitoring.

Results Evidence



Performance Analytics

Calculated dynamically from real database records

Refresh DB

Database Performance Analytics

Calculated dynamically from real database records.

Task Completion Status Breakdown

Completed Tasks Pending Tasks



Tasks Priority Distribution

Task Count



4.5 Data Analysis & Engineering Judgment

The system output is determined by the defined decision rules and the values entered by the user. The use of explicit IF-ELSE conditions makes the decision process transparent and easy to verify. The Priority Queue ensures that tasks assigned higher priority are considered before lower-priority tasks. An important engineering decision was to avoid displaying fabricated dashboard values. This ensures that task counts and farm records represent actual system data. Since the current system uses predefined decision rules, the project should not claim machine-learning accuracy or predictive performance without an independently trained and tested model.

4.6 Comparison with Existing Solutions & Achievement of Objectives

Objective	Evidence	Achievement
-----------	----------	-------------

Design agriculture management system	Overall system architecture and modules	[Fully]
Develop web interface	Implemented website	[Fully]
Analyse farm conditions	Module 2 IF-ELSE logic	[Fully]
Generate farm tasks	Condition-based task generation	[Fully]
Prioritize tasks	Module 1 Priority Queue	[Fully]
Track task completion	Module 3 task status tracking	[Fully]
Monitor performance	Dashboard/task counts	[Fully]
Avoid fabricated values	Dynamic user-entered information	[Fully]

4.7 Strengths & Limitations Strengths

- Simple and understandable user interface.
- Modular system design.
- Uses Priority Queue for task prioritization.
- Uses transparent IF-ELSE decision logic.
- Provides task completion monitoring.
- Supports organized farm-data management.
- Avoids fabricated dashboard information.
- Provides a foundation for future AI and IoT integration.

Limitations

- Current decision support is based on predefined rules.
- The system depends on user-entered information.
- Real-time IoT sensor integration is not included in the current prototype.
- Real-time weather API integration is not included unless separately implemented.
- No machine-learning accuracy should be claimed without actual model training and validation.
- The system does not directly control irrigation or other physical agricultural equipment.

4.8 Project Planning, Roles & Budget Project Milestones

Phase	Activity
Phase 1	Problem identification and requirement analysis
Phase 2	Literature review
Phase 3	System design
Phase 4	Module development

Phase 5	Website development
Phase 6	Module integration
Phase 7	Testing and debugging
Phase 8	Documentation and presentation

Team Roles

Student	Assigned Responsibility	Actual Contribution
R Geethika – 192425101	Module 1	Priority-Based Farm Task Management
B Nikhila – 192472102	Module 3	Task Completion and Performance Monitoring
Y Divya Sree – 192472357	Module 2 + Website + PPT	Conditional Decision Logic-Based Farm Condition Analysis, website development and PPT preparation

CHAPTER 5

CONCLUSION, FUTURE WORK AND REFLECTION

5.1 Conclusion

The AI-Based Sustainable Agriculture Management System was developed to provide software-based decision support for agricultural task management. The system combines farm-condition analysis, priority-based task management, and task completion monitoring.

Module 1 uses a Priority Queue to arrange tasks according to their importance. Module 2 uses IF-ELSE decision logic to analyse soil moisture, crop health, and weather conditions and generate appropriate tasks. Module 3 tracks task completion and provides information about pending and completed tasks.

The project demonstrates that a modular software system can organize farm information and support structured agricultural decision making. The current system is intentionally presented as a decision-support prototype based on defined rules and does not claim trained machine learning accuracy. The system also emphasizes the use of actual user-entered information rather than fabricated dashboard values.

5.2 Future Work

The following improvements can be considered for future development:

- Integration of IoT sensors for real-time soil and environmental data.
- Integration of real-time weather information.
- Development of machine-learning models for crop and disease prediction.
- Image-based crop disease detection.
- Advanced crop recommendation systems.
- Automated irrigation monitoring and control.
- Mobile application development.
- Multilingual support for farmers.
- Cloud-based data storage and remote access.
- Advanced analytics and visualization.
- Integration of historical farm data for predictive decision support.

5.3 Professional Learning & Individual Reflection

New Knowledge Acquired

- Understanding of software-based agricultural decision support.
- Practical use of IF-ELSE decision logic.
- Understanding of Priority Queue based task management.
- Website development and interface design.

- Understanding of database-based information management.
- Technical documentation and presentation preparation.

Engineering & Professional Skills Developed

- Problem analysis.
- Modular software design.
- Web development.
- Algorithm implementation.
- Debugging and testing.
- Team coordination.
- Technical documentation.
- Technical presentation.

Individual Reflection – Y Divya Sree

My main contribution to the project was website development, Module 2 implementation, and PPT preparation. The most difficult part was connecting the farm-condition inputs with the appropriate decisions while keeping the system simple and understandable. I solved this by organizing the conditions into clear IF-ELSE rules and connecting them with the required task generation. I independently developed my understanding of the website interface and decision-logic implementation. If I redesigned the project, I would further improve the interface and integrate real-time weather and IoT data.

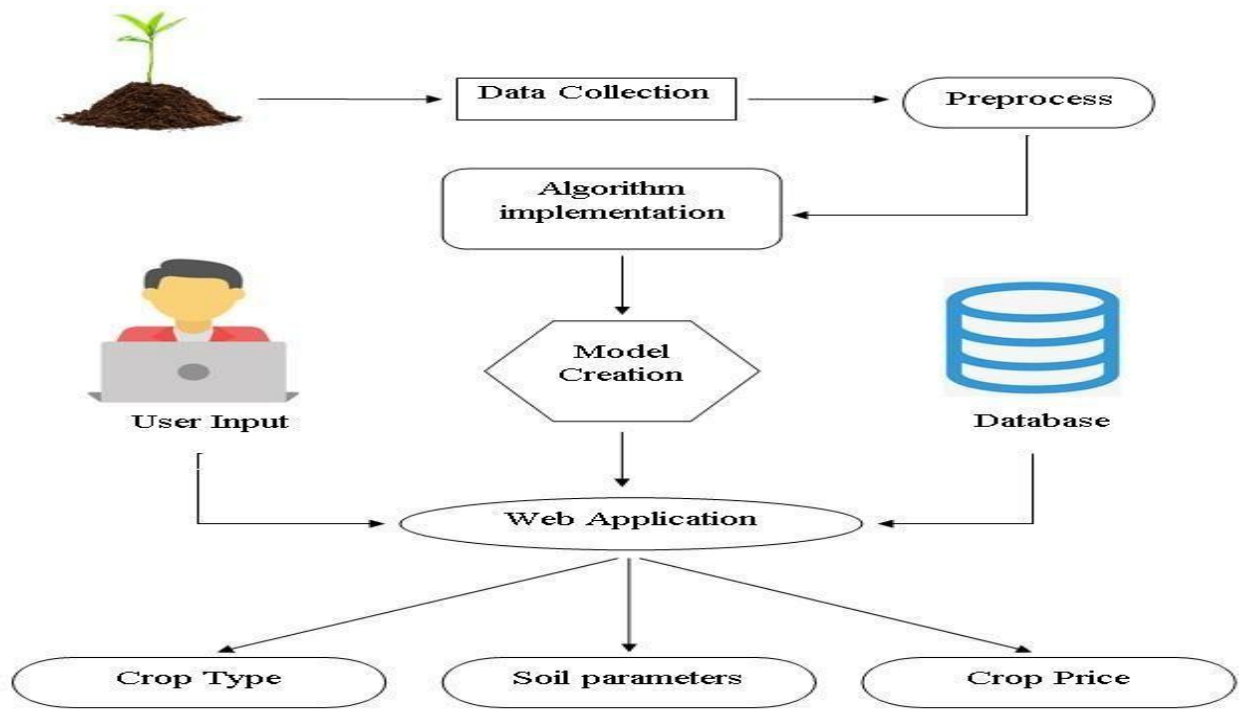
REFERENCES

- [1] A. Sabharwal and S. R. N. Reddy, “Advancing Sustainable Agriculture: Integrating AI and IoT for Soil Nutrient Management—A Systematic Review,” *Soil Use and Management*, Wiley, 2026.
- [2] A. K. M. Pervez, M. S. Kabir, F. A. Prodhan, *et al.*, “Mapping Recent Trends in Artificial Intelligence Research for Sustainable Agriculture: A Bibliometric and Systematic Review,” *Discover Artificial Intelligence*, Springer, 2026.
- [3] Z. Ali, A. Muhammad, N. Lee, M. Waqar, and S. W. Lee, “Artificial Intelligence for Sustainable Agriculture: A Comprehensive Review of AI-Driven Technologies in Crop Production,” *Sustainability*, vol. 17, no. 5, p. 2281, 2025.
- [4] V. Subashini, N. Sriram, and C. Karthikeyan, “Application of Artificial Intelligence in Agriculture and Allied Sectors: A Comprehensive Review Towards Sustainable Solutions,” *Plant Science Today*, 2025.
- [5] Aashu, K. Rajwar, M. Pant, and K. Deep, “Application of Machine Learning in Agriculture: Recent Trends and Future Research Avenues,” *arXiv*, 2024.
- [6] Oracle, “Java Documentation,” Oracle.
- [7] OWASP Foundation, “OWASP Top 10 Web Application Security Risks.”
- [8] International Organization for Standardization, “ISO/IEC 25010: Systems and Software Engineering — Systems and Software Quality Requirements and Evaluation (SQuaRE) — Product Quality Model.”
- [9] World Wide Web Consortium, “Web Content Accessibility Guidelines (WCAG) 2.2.”

Note: Add the actual URLs, access dates, software versions, datasets, repositories, and AI-assisted development resources that were actually used in your project. Do not add resources that you did not use.

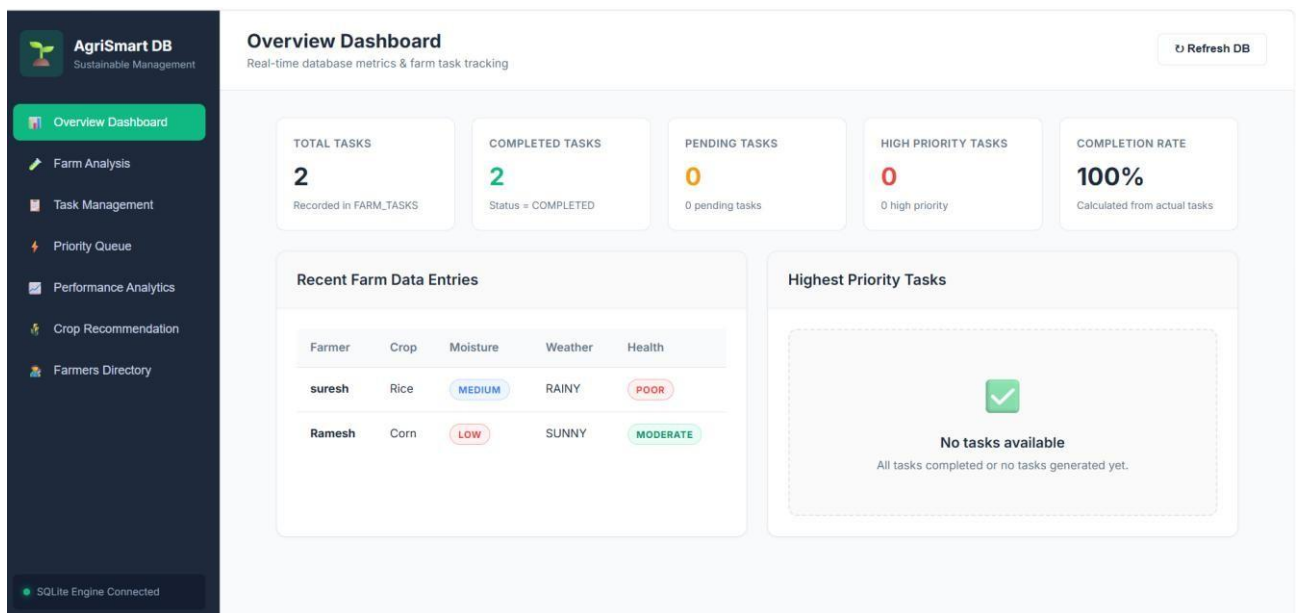
APPENDICES

Appendix A – System Architecture and Workflow



User Input → Farm Condition Analysis → Task Generation → Priority Queue → Task Status → Dashboard

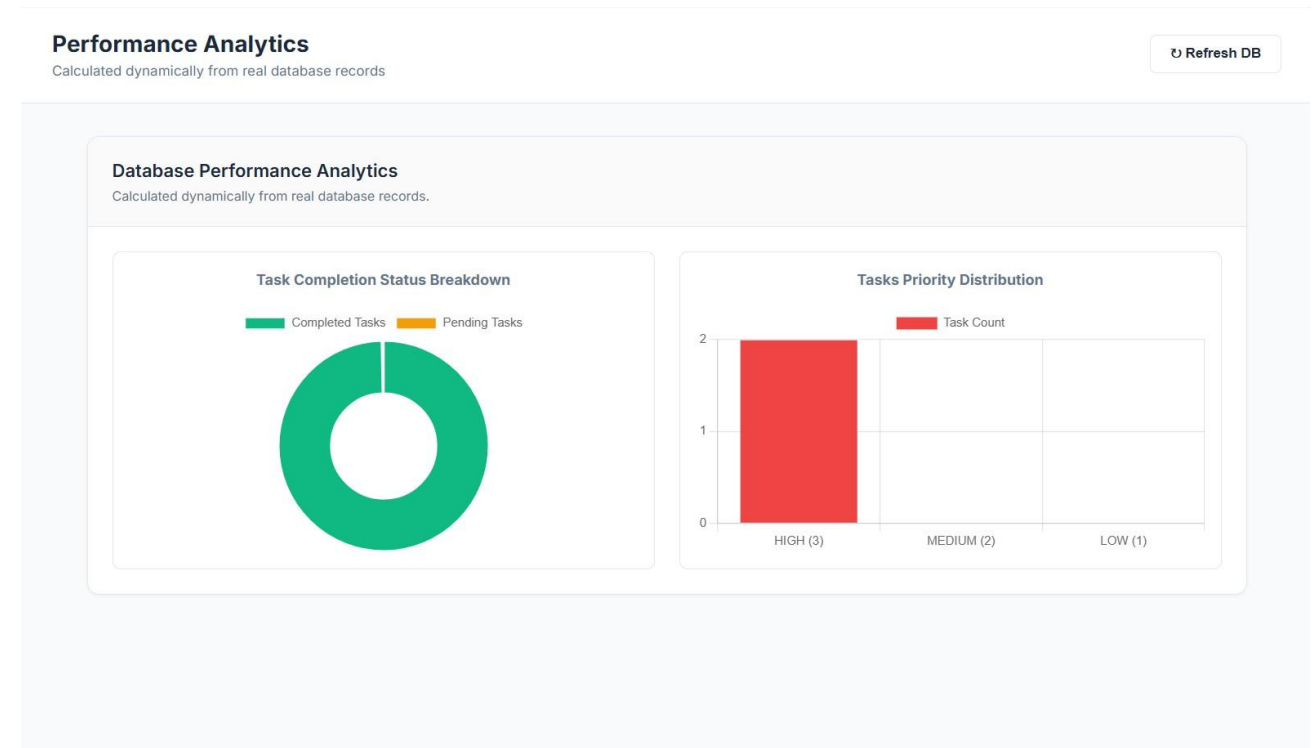
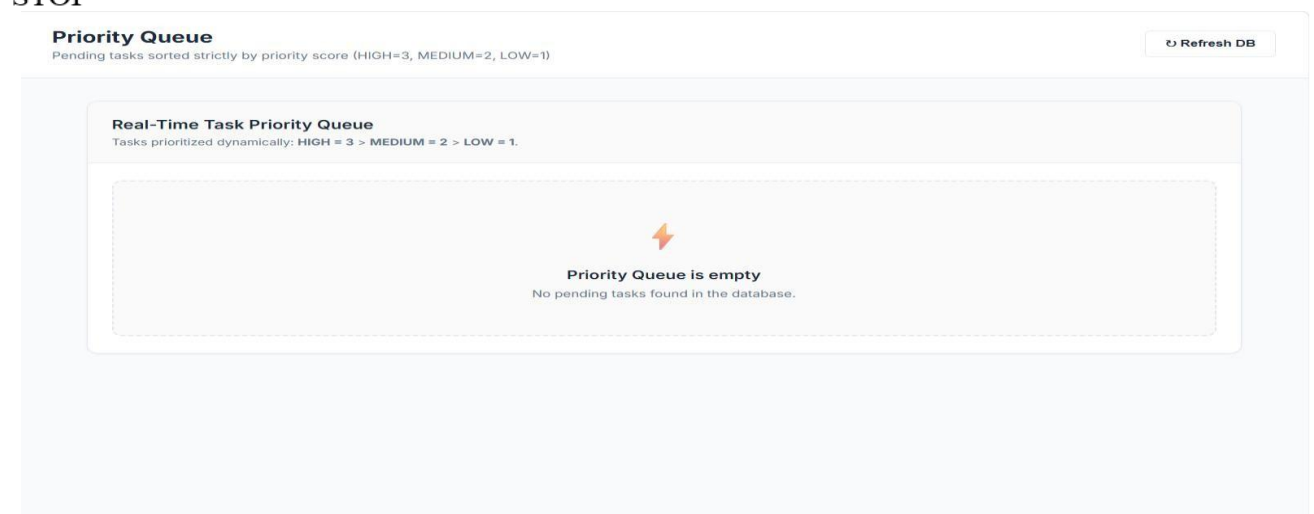
Appendix B – Source Code / Implementation Evidence



```

START
Read crop, soil type, soil moisture,
weather and crop health
IF soil moisture is LOW
    Generate Irrigation Task
    Assign HIGH priority
END IF
IF crop health is POOR
    Generate Disease Monitoring Task
    Assign HIGH priority
END IF
IF weather is RAINY
    Generate Disease Monitoring Task
END IF
Send generated tasks to Priority Queue
STOP

```



Farmer Name *

Crop Name *

Soil Type *
-- Select Soil Type --

Soil Moisture Level *
-- Select Soil Moisture --

Weather Conditions *
-- Select Weather --

Crop Health Status *
-- Select Crop Health --

Save to Database & Run Module 2 Analysis

Clear Form

Data Saved to SQLite & Module 2 Analyzed
Farm Data ID #3 recorded for farmer "adhi".

Generated Tasks (2):

Irrigation Task
HIGH (3)
Rule Reason: Soil moisture is LOW
PENDING

Disease Monitoring Task
HIGH (3)
Rule Reason: Weather is RAINY
PENDING

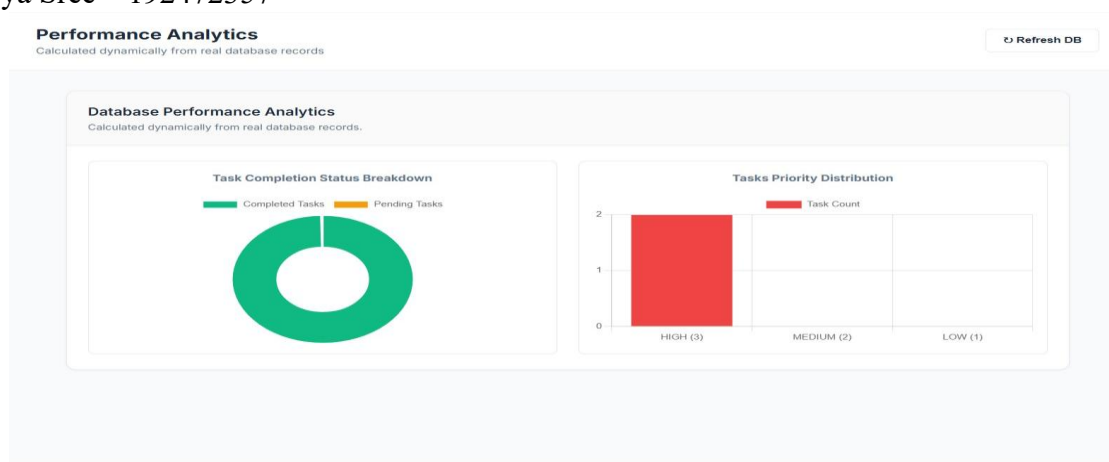
Real-Time Task Priority Queue
Tasks prioritized dynamically: HIGH = 3 > MEDIUM = 2 > LOW = 1.

Showing 2 pending task(s) ordered by highest priority value first.

#1	Irrigation Task HIGH (PRIORITY SCORE: 3) Reason: Soil moisture is LOW Farmer: adhi Crop: tomato	Mark Completed
#2	Disease Monitoring Task HIGH (PRIORITY SCORE: 3) Reason: Weather is RAINY Farmer: adhi Crop: tomato	Mark Completed

Appendix D – Individual Contribution Evidence

Y Divya Sree – 192472357



view, complete, and manage tasks generated in SWLITE database

Database Task Records

Tasks generated from actual user-entered farm conditions.

All Statuses

All Priorities

Irrigation Task

HIGH (SCORE: 3)

COMPLETED

Reason: Soil moisture is LOW

Farmer: Ramesh Crop: Corn Created: 1/9/2026, 6:01:59 am Completed: 1/9/2026, 1:03:45 pm

Delete

Disease Monitoring Task

HIGH (SCORE: 3)

COMPLETED

Reason: Crop health is POOR & Weather is RAINY

Farmer: suresh Crop: Rice Created: 1/9/2026, 6:03:46 am Completed: 1/9/2026, 1:03:47 pm

Delete

Irrigation Task

HIGH (SCORE: 3)

PENDING

Reason: Soil moisture is LOW

Farmer: adhi Crop: tomato Created: 2/9/2026, 11:38:05 am

Complete Task

Delete

Disease Monitoring Task

HIGH (SCORE: 3)

PENDING

Reason: Weather is RAINY

Farmer: adhi Crop: tomato Created: 2/9/2026, 11:38:05 am

Complete Task

Delete

32